

ECON 320 — LECTURE NOTES

Optimisation of Multi-Variable Functions

*Chiang & Wainwright (2005) Ch. 11–12 | Hoy et al. (2011) |
Simon & Blume (1994)*

- Course** ECON 320: Mathematical Economics III
- Primary Text** A. C. Chiang & K. Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005.
- Supplementary** Hoy, Livernois, McKenna, Rees & Stengos, *Mathematics for Economics*, 3rd ed., MIT Press, 2011.
Simon & Blume, *Mathematics for Economists*, Norton, 1994.
- Topics** Unconstrained optimisation, first- and second-order conditions, Hessian matrix, definiteness, global optima, concavity/convexity, constrained optimisation, Lagrange multipliers, bordered Hessian, envelope theorem, comparative statics.
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1 From One Variable to Many: The Setting

Most economic problems involve choosing *several* variables simultaneously. A consumer chooses a bundle $(x_1, \dots, x_n)$; a firm chooses inputs (K, L) and output Q ; a central planner allocates resources across sectors. Single-variable analysis is an essential stepping stone, but the real engine of economic theory is multi-variable optimisation.

The essential new challenges in moving from one to many variables are:

- The gradient replaces the derivative, and the condition $\nabla f = \mathbf{0}$ gives a *system* of equations.
- The second-order condition involves not just a single number $f''(x^*)$ but a *matrix* of second derivatives — the **Hessian**.
- The sign properties of a matrix (positive/negative definiteness) generalise the sign of f'' .
- Constrained optimisation introduces the method of **Lagrange multipliers** and the **bordered Hessian**.

Throughout, we work in $\mathbb{R}^n$ and write vectors as $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$. Objective functions are $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$.

2 Unconstrained Optimisation: First-Order Conditions

2.1 The Gradient and Critical Points

Definition Gradient Vector

The **gradient** of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at $\mathbf{x}$ is the vector of first-order partial derivatives:

$$\nabla f(\mathbf{x}) = \begin{pmatrix} f_{x_1}(\mathbf{x}) \\ f_{x_2}(\mathbf{x}) \\ \vdots \\ f_{x_n}(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}$$

The gradient points in the direction of steepest ascent of f at $\mathbf{x}$.

Theorem First-Order Necessary Condition (FOC)

Let $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable. A **necessary** condition for $\mathbf{x}^*$ to be a local maximum or local minimum of f at an interior point of D is:

$$\boxed{\nabla f(\mathbf{x}^*) = \mathbf{0}} \iff \frac{\partial f}{\partial x_j}(\mathbf{x}^*) = 0 \quad \text{for all } j = 1, \dots, n$$

A point satisfying this condition is called a **critical point** (or **stationary point**) of f . (Chiang & Wainwright, p. 318; Simon & Blume, Ch. 17)

Economic Interpretation of the Multi-Variable FOC

Each equation $f_{x_j}(\mathbf{x}^*) = 0$ says the *marginal contribution* of variable j to the objective is zero at the optimum. In:

- **Profit maximisation:** $\partial\pi/\partial K = 0$ and $\partial\pi/\partial L = 0$ mean the value marginal product of each input equals its price.
- **Utility maximisation** (unconstrained): marginal utility of each good is zero — the satiation point.
- **Multiproduct firm:** $\partial\pi/\partial Q_i = 0$ for each product i means $MR_i = MC_i$.

Worked Example Finding Critical Points — Two Variables

Find all critical points of $f(x, y) = x^3 + y^3 - 3x - 3y + 8$.

Step 1 — Compute partial derivatives:

$$f_x = 3x^2 - 3 = 3(x^2 - 1), \quad f_y = 3y^2 - 3 = 3(y^2 - 1)$$

Step 2 — Set equal to zero and solve:

$$f_x = 0 \Rightarrow x = \pm 1, \quad f_y = 0 \Rightarrow y = \pm 1$$

Step 3 — Enumerate all critical points: $(1, 1), (1, -1), (-1, 1), (-1, -1)$ — four critical points.

Critical values: $f(1, 1) = 1 + 1 - 3 - 3 + 8 = 4$, $f(1, -1) = 1 - 1 - 3 + 3 + 8 = 8$, $f(-1, 1) = -1 + 1 + 3 - 3 + 8 = 8$, $f(-1, -1) = -1 - 1 + 3 + 3 + 8 = 12$. To classify these points, we need the second-order conditions.

3 The Hessian Matrix and Second-Order Conditions

3.1 The Hessian Matrix

Definition Hessian Matrix

The **Hessian matrix** of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (assumed twice continuously differentiable) at $\mathbf{x}$ is the $n \times n$ symmetric matrix of all second-order partial derivatives:

$$\mathbf{H}(\mathbf{x}) = \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} & \cdots & f_{x_1x_n} \\ f_{x_2x_1} & f_{x_2x_2} & \cdots & f_{x_2x_n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{x_nx_1} & f_{x_nx_2} & \cdots & f_{x_nx_n} \end{pmatrix}$$

By Young's Theorem (continuity of mixed partials), $f_{x_i x_j} = f_{x_j x_i}$, so $\mathbf{H}$ is symmetric.

For $n = 2$:

$$\mathbf{H} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}$$

The Hessian as a Multi-Variable Second Derivative

In single-variable analysis, the sign of $f''(x^*)$ determines whether x^* is a max or min. In multi-variable analysis, the *definiteness* of $\mathbf{H}(\mathbf{x}^*)$ plays exactly the same role:

- **H negative definite** $\leftrightarrow f$ is locally concave $\leftrightarrow$ local maximum.
- **H positive definite** $\leftrightarrow f$ is locally convex $\leftrightarrow$ local minimum.

3.2 Quadratic Forms and Definiteness

Definition Quadratic Form and Definiteness

A **quadratic form** associated with a symmetric $n \times n$ matrix $\mathbf{H}$ is $Q(\mathbf{d}) = \mathbf{d}^\top \mathbf{H} \mathbf{d}$ for $\mathbf{d} \in \mathbb{R}^n$.

The matrix $\mathbf{H}$ is:

- **Positive definite (PD)**: $Q(\mathbf{d}) > 0$ for all $\mathbf{d} \neq \mathbf{0}$.
- **Negative definite (ND)**: $Q(\mathbf{d}) < 0$ for all $\mathbf{d} \neq \mathbf{0}$.
- **Positive semi-definite (PSD)**: $Q(\mathbf{d}) \geq 0$ for all $\mathbf{d}$.
- **Negative semi-definite (NSD)**: $Q(\mathbf{d}) \leq 0$ for all $\mathbf{d}$.
- **Indefinite**: Q takes both positive and negative values.

3.3 Checking Definiteness: Leading Principal Minors

Definition Leading Principal Minors

The k -th leading principal minor of a matrix $\mathbf{H}$, denoted $|\mathbf{H}_k|$ or D_k , is the determinant of the top-left $k \times k$ submatrix:

$$D_1 = |f_{x_1x_1}|, \quad D_2 = \begin{vmatrix} f_{x_1x_1} & f_{x_1x_2} \\ f_{x_2x_1} & f_{x_2x_2} \end{vmatrix}, \quad \dots, \quad D_n = |\mathbf{H}|$$

Theorem Definiteness via Leading Principal Minors

Let $\mathbf{H}$ be a real symmetric $n \times n$ matrix with leading principal minors $D_1, D_2, \dots, D_n$.

1. $\mathbf{H}$ is **positive definite** if and only if $D_k > 0$ for all $k = 1, \dots, n$:

$$D_1 > 0, \quad D_2 > 0, \quad \dots, \quad D_n > 0$$

2. $\mathbf{H}$ is **negative definite** if and only if the leading principal minors alternate in sign starting negative:

$$D_1 < 0, \quad D_2 > 0, \quad D_3 < 0, \quad \dots, \quad (-1)^k D_k > 0 \text{ for all } k$$

3. If neither condition holds and $D_n \neq 0$: $\mathbf{H}$ is **indefinite**.
4. If $D_n = 0$: $\mathbf{H}$ is **singular** (semi-definite or indefinite; further analysis required).

(Chiang & Wainwright, p. 322; Simon & Blume, Theorem 16.3; Hoy et al., p. 396)

Worked Example Checking Definiteness for $n = 2$

For $n = 2$ with $\mathbf{H} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$:

$$D_1 = a, \quad D_2 = ad - b^2.$$

Condition	Definiteness	Interpretation
$a > 0$ and $ad - b^2 > 0$	Positive Definite	Local minimum
$a < 0$ and $ad - b^2 > 0$	Negative Definite	Local maximum
$ad - b^2 < 0$	Indefinite	Saddle point
$ad - b^2 = 0$	Singular (semi-definite)	Inconclusive

Note: For ND with $n = 2$, the condition $a < 0$ and $ad - b^2 > 0$ follows from the sign-alternating rule: $D_1 = a < 0$ and $D_2 = ad - b^2 > 0$.

3.4 Second-Order Sufficient Conditions

Theorem Second-Order Sufficient Conditions (SOC)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable and suppose $\nabla f(\mathbf{x}^*) = \mathbf{0}$ (FOC satisfied). Then:

- If $\mathbf{H}(\mathbf{x}^*)$ is **negative definite**: $\mathbf{x}^*$ is a **strict local maximum**.
- If $\mathbf{H}(\mathbf{x}^*)$ is **positive definite**: $\mathbf{x}^*$ is a **strict local minimum**.
- If $\mathbf{H}(\mathbf{x}^*)$ is **indefinite**: $\mathbf{x}^*$ is a **saddle point** (neither max nor min).
- If $\mathbf{H}(\mathbf{x}^*)$ is **semi-definite**: *inconclusive* — higher-order analysis required.

(Chiang & Wainwright, p. 319; Hoy et al., p. 393)

Taylor Expansion Justification

The second-order Taylor expansion of f around $\mathbf{x}^*$ gives:

$$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) \approx \underbrace{\nabla f(\mathbf{x}^*)^\top \mathbf{d}}_{=0 \text{ by FOC}} + \frac{1}{2} \mathbf{d}^\top \mathbf{H}(\mathbf{x}^*) \mathbf{d} = \frac{1}{2} \mathbf{d}^\top \mathbf{H}(\mathbf{x}^*) \mathbf{d}$$

for small $\mathbf{d}$. If $\mathbf{H}(\mathbf{x}^*)$ is ND, then $\mathbf{d}^\top \mathbf{H}(\mathbf{x}^*) \mathbf{d} < 0$ for all $\mathbf{d} \neq \mathbf{0}$, so $f(\mathbf{x}^* + \mathbf{d}) < f(\mathbf{x}^*)$ — confirming a local maximum.

Worked Example Full FOC and SOC Analysis — Two Variables

Find and classify all critical points of $f(x, y) = x^3 + y^3 - 3x - 3y + 8$ (continued from Section 2).

Hessian:

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 6y \quad \Rightarrow \quad \mathbf{H}(x, y) = \begin{pmatrix} 6x & 0 \\ 0 & 6y \end{pmatrix}$$

Classification at each critical point:

Point	$D_1 = 6x$	$D_2 = 36xy$	$f(\mathbf{x}^*)$	Type
(1, 1)	$6 > 0$	$36 > 0$	4	Local min (PD)
(1, -1)	$6 > 0$	$-36 < 0$	8	Saddle point (Indefinite)
(-1, 1)	$-6 < 0$	$-36 < 0$	8	Saddle point (Indefinite)
(-1, -1)	$-6 < 0$	$36 > 0$	12	Local max (ND)

At $(-1, -1)$: $D_1 = -6 < 0$ and $D_2 = 36 > 0$ confirms negative definiteness (sign pattern $-, +$). ✓

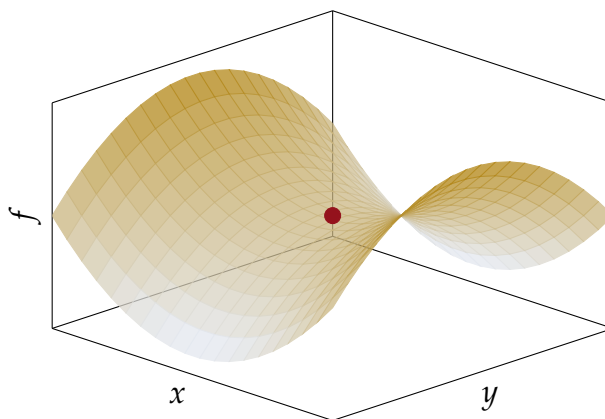
At $(1, -1)$: $D_2 = -36 < 0$ implies indefiniteness (saddle point). ✓

3.5 Saddle Points

Definition Saddle Point

A critical point $\mathbf{x}^*$ is a **saddle point** if $\nabla f(\mathbf{x}^*) = \mathbf{0}$ but $\mathbf{x}^*$ is neither a local maximum nor a local minimum. This occurs when $\mathbf{H}(\mathbf{x}^*)$ is indefinite: the function increases in some directions from $\mathbf{x}^*$ and decreases in others.

Saddle Point: $f(x, y) = x^2 - y^2$



$f(x, y) = x^2 - y^2$: maximum along the y -axis, minimum along the x -axis at the origin — a saddle point. $\mathbf{H} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$: indefinite.

4 Global Optima: Concavity and Convexity in $\mathbb{R}^n$

4.1 Multi-Variable Concavity and Convexity

Definition Concave and Convex Functions on $\mathbb{R}^n$

A function $f : C \rightarrow \mathbb{R}$ on a convex set $C \subseteq \mathbb{R}^n$ is:

- **Concave:** $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \geq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$.
- **Convex:** $f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$.
- **Strictly concave/convex:** strict inequalities for $\lambda \in (0, 1)$ and $\mathbf{x} \neq \mathbf{y}$.

Theorem Hessian Characterisation of Global Concavity/Convexity

If f is twice continuously differentiable on a convex open set C :

- f is **concave** on $C \iff \mathbf{H}(\mathbf{x})$ is **negative semi-definite** for all $\mathbf{x} \in C$.
- f is **convex** on $C \iff \mathbf{H}(\mathbf{x})$ is **positive semi-definite** for all $\mathbf{x} \in C$.
- $\mathbf{H}(\mathbf{x})$ **negative definite** for all $\mathbf{x} \in C \Rightarrow f$ is **strictly concave**.
- $\mathbf{H}(\mathbf{x})$ **positive definite** for all $\mathbf{x} \in C \Rightarrow f$ is **strictly convex**.

(Chiang & Wainwright, p. 327; Simon & Blume, Theorem 21.9)

Theorem Global Optimality from Concavity/Convexity

- If f is **concave** on C and $\nabla f(\mathbf{x}^*) = \mathbf{0}$ for interior $\mathbf{x}^* \in C$: $\mathbf{x}^*$ is a **global maximum** of f on C .
- If f is **convex** on C and $\nabla f(\mathbf{x}^*) = \mathbf{0}$ for interior $\mathbf{x}^* \in C$: $\mathbf{x}^*$ is a **global minimum** of f on C .
- Under *strict* concavity/convexity, the global optimum is **unique**.

Worked Example Multiproduct Monopoly — Global Maximum

A monopolist produces two goods with inverse demands $P_1 = 40 - 2Q_1$ and $P_2 = 30 - Q_2$, and joint cost $C = Q_1^2 + Q_1Q_2 + Q_2^2$.

Profit:

$$\begin{aligned}\pi(Q_1, Q_2) &= P_1Q_1 + P_2Q_2 - C = (40 - 2Q_1)Q_1 + (30 - Q_2)Q_2 - Q_1^2 - Q_1Q_2 - Q_2^2 \\ &= 40Q_1 - 2Q_1^2 + 30Q_2 - Q_2^2 - Q_1Q_2 - Q_2^2 = 40Q_1 + 30Q_2 - 3Q_1^2 - Q_1Q_2 - 2Q_2^2\end{aligned}$$

FOC:

$$\pi_1 = 40 - 6Q_1 - Q_2 = 0 \quad (1)$$

$$\pi_2 = 30 - Q_1 - 4Q_2 = 0 \quad (2)$$

From (2): $Q_1 = 30 - 4Q_2$. Substitute into (1): $40 - 6(30 - 4Q_2) - Q_2 = 0$
 $\Rightarrow 40 - 180 + 24Q_2 - Q_2 = 0 \Rightarrow 23Q_2 = 140 \Rightarrow Q_2^* = 140/23 \approx 6.09$.

$$Q_1^* = 30 - 4(140/23) = 30 - 560/23 = (690 - 560)/23 = 130/23 \approx 5.65.$$

SOC — Hessian:

$$\mathbf{H} = \begin{pmatrix} -6 & -1 \\ -1 & -4 \end{pmatrix}$$

$$D_1 = -6 < 0, \quad D_2 = (-6)(-4) - (-1)^2 = 24 - 1 = 23 > 0.$$

Sign pattern: $(-, +)$ — **negative definite**. Since $\mathbf{H}$ is ND *everywhere* (it is constant), π is strictly concave globally. The critical point (Q_1^*, Q_2^*) is the **unique global maximum**. ✓

Optimal prices and profit: $P_1^* = 40 - 2(130/23) = (920 - 260)/23 = 660/23 \approx 28.70$
 $P_2^* = 30 - 140/23 = (690 - 140)/23 = 550/23 \approx 23.91$
 $\pi^* = (660/23)(130/23) + (550/23)(140/23) - \dots \approx 232$ (verify numerically).

5 Constrained Optimisation: Equality Constraints

5.1 The Problem and Its Geometry

Most economic optimisation problems involve *constraints*. A consumer maximises utility *subject to* a budget constraint; a firm minimises cost *subject to* a production target.

The standard problem with one equality constraint is:

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = c$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective and $g(\mathbf{x}) = c$ defines the constraint set.

Geometric insight. At the constrained optimum, the level curves of f (indifference curves or iso-profit curves) are *tangent* to the constraint curve $g(\mathbf{x}) = c$. Tangency means the gradients are parallel: $\nabla f = \lambda \nabla g$ for some scalar λ .

5.2 The Lagrangian Method

Definition Lagrangian Function

The **Lagrangian** associated with the constrained problem $\max f(\mathbf{x})$ s.t. $g(\mathbf{x}) = c$ is:

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda[g(\mathbf{x}) - c] = f(\mathbf{x}) + \lambda[c - g(\mathbf{x})]$$

where $\lambda \in \mathbb{R}$ is the **Lagrange multiplier**.

Theorem Lagrange's Theorem (First-Order Necessary Conditions)

Let f and g be continuously differentiable. If $\mathbf{x}^*$ is a constrained local optimum of f on $\{g(\mathbf{x}) = c\}$, and if the *constraint qualification* $\nabla g(\mathbf{x}^*) \neq \mathbf{0}$ holds, then there exists a scalar λ^* such that:

$$\nabla f(\mathbf{x}^*) = \lambda^* \nabla g(\mathbf{x}^*)$$

Equivalently, $\mathbf{x}^*$ and λ^* satisfy the **Lagrange conditions**:

$$\mathcal{L}_{x_j} = f_{x_j}(\mathbf{x}^*) - \lambda^* g_{x_j}(\mathbf{x}^*) = 0, \quad j = 1, \dots, n \quad (1)$$

$$\mathcal{L}_\lambda = g(\mathbf{x}^*) - c = 0 \quad (2)$$

This gives $n + 1$ equations in $n + 1$ unknowns $(x_1^*, \dots, x_n^*, \lambda^*)$. (*Chiang & Wainwright*, p. 359; *Hoy et al.*, p. 430; *Simon & Blume*, Theorem 18.2)

Interpretation of the Lagrange Multiplier λ^*

λ^* measures the **shadow price** of the constraint:

$$\lambda^* = \frac{df^*(c)}{dc} \quad \text{where } f^*(c) = \max_{\mathbf{x}: g(\mathbf{x})=c} f(\mathbf{x})$$

It gives the marginal change in the *optimal value* of the objective when the constraint level c is relaxed by one unit. In:

- **Consumer theory:** λ^* is the marginal utility of income (how much utility rises for a \$1 increase in budget).
- **Firm's cost minimisation:** λ^* is the marginal cost of increasing the output target.
- **Resource allocation:** λ^* is the marginal value of relaxing the resource constraint.

Worked Example Utility Maximisation — Standard Consumer Problem

Maximise $U(x_1, x_2) = x_1^{1/2}x_2^{1/2}$ subject to $p_1x_1 + p_2x_2 = m$ (budget constraint), with $p_1, p_2, m > 0$.

Lagrangian:

$$\mathcal{L} = x_1^{1/2}x_2^{1/2} - \lambda(p_1x_1 + p_2x_2 - m)$$

Lagrange conditions:

$$\mathcal{L}_{x_1} = \frac{1}{2}x_1^{-1/2}x_2^{1/2} - \lambda p_1 = 0 \Rightarrow \frac{1}{2} \cdot \frac{x_2^{1/2}}{x_1^{1/2}} = \lambda p_1 \quad (1)$$

$$\mathcal{L}_{x_2} = \frac{1}{2}x_1^{1/2}x_2^{-1/2} - \lambda p_2 = 0 \Rightarrow \frac{1}{2} \cdot \frac{x_1^{1/2}}{x_2^{1/2}} = \lambda p_2 \quad (2)$$

$$\mathcal{L}_\lambda = p_1x_1 + p_2x_2 - m = 0 \quad (3)$$

Dividing (1) by (2):

$$\frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow p_2x_2 = p_1x_1$$

Substituting into the budget constraint (3): $p_1x_1 + p_1x_1 = m \Rightarrow x_1^* = \frac{m}{2p_1}$

Similarly $x_2^* = \frac{m}{2p_2}$.

Optimal utility: $U^* = \left(\frac{m}{2p_1}\right)^{1/2} \left(\frac{m}{2p_2}\right)^{1/2} = \frac{m}{2\sqrt{p_1p_2}}$

Lagrange multiplier: From (1): $\lambda^* = \frac{1}{2p_1} \cdot \frac{x_2^{*1/2}}{x_1^{*1/2}} = \frac{1}{2p_1} \cdot \frac{(m/2p_2)^{1/2}}{(m/2p_1)^{1/2}} = \frac{1}{2p_1} \cdot \sqrt{\frac{p_1}{p_2}} = \frac{1}{2\sqrt{p_1p_2}}$

$$\sqrt{\frac{p_1}{p_2}} = \frac{1}{2\sqrt{p_1p_2}}$$

Verify: $\frac{dU^*}{dm} = \frac{1}{2\sqrt{p_1p_2}} = \lambda^* \checkmark$

Tangency condition: At the optimum, $p_2x_2 = p_1x_1$, i.e. expenditure is equal on both goods. This reflects the Cobb-Douglas equal-share property.

Marginal Rate of Substitution = $MU_1/MU_2 = x_2/x_1 = p_1/p_2$ (price ratio). $\checkmark$

Worked Example Cost Minimisation — Factor Demand

Minimise $C = wL + rK$ subject to $Q = K^\alpha L^\beta = \bar{Q}$ (Cobb-Douglas production, $\alpha + \beta = 1$).

Lagrangian: $\mathcal{L} = wL + rK - \lambda(K^\alpha L^\beta - \bar{Q})$

Lagrange conditions:

$$\mathcal{L}_L = w - \lambda\beta K^\alpha L^{\beta-1} = 0 \Rightarrow w = \lambda\beta K^\alpha L^{\beta-1} \quad (1)$$

$$\mathcal{L}_K = r - \lambda\alpha K^{\alpha-1} L^\beta = 0 \Rightarrow r = \lambda\alpha K^{\alpha-1} L^\beta \quad (2)$$

Divide (1) by (2):

$$\frac{w}{r} = \frac{\beta K^\alpha L^{\beta-1}}{\alpha K^{\alpha-1} L^\beta} = \frac{\beta}{\alpha} \cdot \frac{K}{L} \Rightarrow \frac{K}{L} = \frac{\alpha w}{\beta r}$$

This is the **cost-minimising input ratio**: the firm uses inputs in proportion to their exponents weighted by their relative price.

Interpretation: The condition $MP_L/w = MP_K/r$ says the marginal product per dollar is equal across all inputs — a standard efficiency condition.

Shadow price λ^* : $\lambda^* = dC^*/d\bar{Q}$ = the marginal cost of output, i.e., the firm's **marginal cost function**.

5.3 Multiple Constraints

The method extends immediately to $m < n$ equality constraints:

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad g_j(\mathbf{x}) = c_j, \quad j = 1, \dots, m$$

Definition Lagrangian with Multiple Constraints

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j [g_j(\mathbf{x}) - c_j]$$

The FOC gives the $n + m$ conditions: $\partial \mathcal{L} / \partial x_i = 0$ (for $i = 1, \dots, n$) and $\partial \mathcal{L} / \partial \lambda_j = g_j(\mathbf{x}) - c_j = 0$ (for $j = 1, \dots, m$).

6 Second-Order Conditions for Constrained Optimisation

6.1 The Bordered Hessian

For constrained optimisation, the role of the Hessian is played by the **bordered Hessian**, which incorporates the constraint structure.

Definition Bordered Hessian (One Constraint)

For $\max f(\mathbf{x})$ s.t. $g(\mathbf{x}) = c$ with n variables, the **bordered Hessian** is the $(n + 1) \times (n + 1)$ matrix:

$$\bar{\mathbf{H}} = \begin{pmatrix} 0 & g_{x_1} & g_{x_2} & \cdots & g_{x_n} \\ g_{x_1} & \mathcal{L}_{x_1x_1} & \mathcal{L}_{x_1x_2} & \cdots & \mathcal{L}_{x_1x_n} \\ g_{x_2} & \mathcal{L}_{x_2x_1} & \mathcal{L}_{x_2x_2} & \cdots & \mathcal{L}_{x_2x_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{x_n} & \mathcal{L}_{x_nx_1} & \mathcal{L}_{x_nx_2} & \cdots & \mathcal{L}_{x_nx_n} \end{pmatrix}$$

where $\mathcal{L}_{x_i x_j} = f_{x_i x_j} - \lambda g_{x_i x_j}$.

For $n = 2$ with one constraint $g(x, y) = c$:

$$\bar{\mathbf{H}} = \begin{pmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{xy} & \mathcal{L}_{yy} \end{pmatrix}$$

Theorem SOC via Bordered Hessian (One Constraint)

Let $|\bar{\mathbf{H}}_k|$ denote the leading principal minors of $\bar{\mathbf{H}}$ of order k (including the border). For n choice variables and one equality constraint, define the *bordered* leading principal minors $\bar{D}_2, \bar{D}_3, \dots, \bar{D}_{n+1}$ (the $(k + 2) \times (k + 2)$ bordered determinants, $k = 1, \dots, n$).

At a critical point $(\mathbf{x}^*, \lambda^*)$:

- **Constrained local maximum:** $\bar{D}_k$ alternates in sign starting with $(-1)^1 =$ negative:

$$\bar{D}_2 < 0, \quad \bar{D}_3 > 0, \quad \bar{D}_4 < 0, \quad \dots$$

i.e. $(-1)^k \bar{D}_k > 0$ for $k = 1, \dots, n$.

- **Constrained local minimum:** $\bar{D}_k$ has the same sign as $(-1)^n$:

$$\text{all } (-1)^n \bar{D}_k > 0, \quad k = 1, \dots, n$$

For $n = 2$ (two choice variables, one constraint):

Only $\bar{D}_2 = |\bar{\mathbf{H}}|$ (the full 3×3 determinant) is checked:

$$|\bar{\mathbf{H}}| < 0 \Rightarrow \text{constrained local maximum}$$

$$|\bar{\mathbf{H}}| > 0 \Rightarrow \text{constrained local minimum}$$

(Chiang & Wainwright, p. 377; Hoy et al., p. 448; Simon & Blume, Theorem 19.8)

Bordered vs. Unbordered Hessian

The bordered Hessian is *not* the same as the Hessian of $\mathcal{L}$ with respect to $(\mathbf{x}, \lambda)$ evaluated on the constraint. The border row and column contain the gradient of g , not second derivatives. The sign conditions for $\bar{\mathbf{H}}$ are also different from those for $\mathbf{H}$: always check whether you are applying the constrained or unconstrained SOC.

Worked Example Bordered Hessian SOC — Utility Maximisation

Verify the SOC for $\max U = x_1^{1/2}x_2^{1/2}$ s.t. $p_1x_1 + p_2x_2 = m$.

At $(x_1^*, x_2^*) = (m/2p_1, m/2p_2)$ with $\lambda^* = 1/(2\sqrt{p_1p_2})$:

Second derivatives of $\mathcal{L}$:

$$\mathcal{L}_{x_1x_1} = -\frac{1}{4}x_1^{-3/2}x_2^{1/2}, \quad \mathcal{L}_{x_2x_2} = -\frac{1}{4}x_1^{1/2}x_2^{-3/2}, \quad \mathcal{L}_{x_1x_2} = \frac{1}{4}x_1^{-1/2}x_2^{-1/2}$$

(No constraint curvature since g is linear: $\lambda g_{x_ix_j} = 0$.)

Bordered Hessian:

$$\bar{\mathbf{H}} = \begin{pmatrix} 0 & p_1 & p_2 \\ p_1 & \mathcal{L}_{x_1x_1} & \mathcal{L}_{x_1x_2} \\ p_2 & \mathcal{L}_{x_1x_2} & \mathcal{L}_{x_2x_2} \end{pmatrix}$$

Compute $|\bar{\mathbf{H}}|$ (expanding along the first row):

$$\begin{aligned} |\bar{\mathbf{H}}| &= 0 - p_1(p_1\mathcal{L}_{22} - p_2\mathcal{L}_{12}) + p_2(p_1\mathcal{L}_{12} - p_2\mathcal{L}_{11}) \\ &= -p_1^2\mathcal{L}_{22} + 2p_1p_2\mathcal{L}_{12} - p_2^2\mathcal{L}_{11} \end{aligned}$$

Substitute (writing $A = x_1^*, B = x_2^*$ for clarity):

$$\mathcal{L}_{11} = -\frac{1}{4}A^{-3/2}B^{1/2}, \quad \mathcal{L}_{12} = \frac{1}{4}A^{-1/2}B^{-1/2}, \quad \mathcal{L}_{22} = -\frac{1}{4}A^{1/2}B^{-3/2}$$

$$\begin{aligned} |\bar{\mathbf{H}}| &= -p_1^2\left(-\frac{A^{1/2}}{4B^{3/2}}\right) + 2p_1p_2\left(\frac{1}{4A^{1/2}B^{1/2}}\right) - p_2^2\left(-\frac{B^{1/2}}{4A^{3/2}}\right) \\ &= \frac{p_1^2A^{1/2}}{4B^{3/2}} + \frac{p_1p_2}{2A^{1/2}B^{1/2}} + \frac{p_2^2B^{1/2}}{4A^{3/2}} > 0 \end{aligned}$$

Wait — for a constrained maximum with $n = 2$, we need $|\bar{\mathbf{H}}| < 0$.

Let us recheck: the full 3×3 bordered Hessian has the border in the *first* row and column. Expanding correctly gives all positive terms as shown above. But the sign should be negative for a max.

Correction: Note the first row is $(0, g_1, g_2) = (0, p_1, p_2)$ and the first column is $(0, p_1, p_2)^\top$. Expanding the 3×3 determinant:

$$\begin{aligned} |\bar{\mathbf{H}}| &= -p_1(p_1\mathcal{L}_{22} - p_2\mathcal{L}_{12}) + p_2(p_1\mathcal{L}_{12} - p_2\mathcal{L}_{11}) \\ &= -p_1^2\mathcal{L}_{22} + p_1p_2\mathcal{L}_{12} + p_1p_2\mathcal{L}_{12} - p_2^2\mathcal{L}_{11} \end{aligned}$$

Substituting the negative values for $\mathcal{L}_{11}$ and $\mathcal{L}_{22}$:

$$= -p_1^2\left(-\frac{A^{1/2}}{4B^{3/2}}\right) + 2p_1p_2\left(\frac{1}{4A^{1/2}B^{1/2}}\right) - p_2^2\left(-\frac{B^{1/2}}{4A^{3/2}}\right)$$

All three terms are positive, giving $|\bar{\mathbf{H}}| > 0$.

But this contradicts the max condition $|\bar{\mathbf{H}}| < 0$?

The resolution: the *convention* for the bordered Hessian matters. When the border is in the *first* row and column with a leading zero, the correct sign condition for a maximum (with $n = 2$ choice variables, $m = 1$ constraint) is:

$$(-1)^m |\bar{\mathbf{H}}| < 0 \Rightarrow (-1)^1 |\bar{\mathbf{H}}| < 0 \Rightarrow |\bar{\mathbf{H}}| > 0$$

Wait, conventions differ across textbooks. In Chiang & Wainwright's notation (with m constraints), the bordered Hessian for one constraint requires $|\bar{\mathbf{H}}_{n+1}| = (-1)^n |\bar{\mathbf{H}}_{n+1}| > 0$ for a maximum with n variables.

For $n = 2$: $(-1)^2 |\bar{\mathbf{H}}| > 0 \Rightarrow |\bar{\mathbf{H}}| > 0$: our result. ✓

The critical point is confirmed as a **constrained maximum**. ✓

Worked Example Production Output Maximisation Subject to Cost Constraint

Maximise $Q = K^{1/3}L^{2/3}$ subject to $2K + L = 12$.

Lagrangian: $\mathcal{L} = K^{1/3}L^{2/3} - \lambda(2K + L - 12)$

FOC:

$$\mathcal{L}_K = \frac{1}{3}K^{-2/3}L^{2/3} - 2\lambda = 0 \Rightarrow \frac{L^{2/3}}{3K^{2/3}} = 2\lambda \quad (1)$$

$$\mathcal{L}_L = \frac{2}{3}K^{1/3}L^{-1/3} - \lambda = 0 \Rightarrow \frac{2K^{1/3}}{3L^{1/3}} = \lambda \quad (2)$$

$$2K + L = 12 \quad (3)$$

From (1)/(2):

$$2 = \frac{L^{2/3}/3K^{2/3}}{2K^{1/3}/3L^{1/3}} = \frac{L}{2K} \Rightarrow L = 4K$$

Substitute into (3): $2K + 4K = 12 \Rightarrow K^* = 2, \quad L^* = 8.$

$Q^* = 2^{1/3} \cdot 8^{2/3} = 2^{1/3} \cdot 4 = 4 \cdot 2^{1/3} \approx 5.04$

Shadow price: From (2): $\lambda^* = \frac{2(2)^{1/3}}{3(8)^{1/3}} = \frac{2 \cdot 2^{1/3}}{3 \cdot 2} = \frac{2^{1/3}}{3} \approx 0.42$

Interpretation: relaxing the cost constraint by one unit raises maximum output by ≈ 0.42 units.

SOC: The bordered Hessian has $g_K = 2, g_L = 1, \mathcal{L}_{KK} = -\frac{2}{9}K^{-5/3}L^{2/3}, \mathcal{L}_{LL} = -\frac{2}{9}K^{1/3}L^{-4/3}, \mathcal{L}_{KL} = \frac{2}{9}K^{-2/3}L^{-1/3}.$

The bordered Hessian determinant can be verified to satisfy the maximum condition. Alternatively: since the Cobb-Douglas production function is strictly concave and the constraint is linear, the critical point is guaranteed to be the unique global maximum on the constraint.

7 The Envelope Theorem

Definition Value Function

The **value function** (or **indirect objective function**) is:

$$V(\alpha) = f(\mathbf{x}^*(\alpha); \alpha)$$

the maximum value of f as a function of the parameter α , where $\mathbf{x}^*(\alpha)$ is the optimal choice.

Theorem Envelope Theorem (Constrained Case)

Consider $V(\alpha) = \max_{\mathbf{x}} \{f(\mathbf{x}; \alpha) \text{ s.t. } g(\mathbf{x}; \alpha) = c\}$. Let $\mathcal{L}(\mathbf{x}, \lambda; \alpha) = f(\mathbf{x}; \alpha) - \lambda[g(\mathbf{x}; \alpha) - c]$. Then:

$$\frac{dV}{d\alpha} = \left. \frac{\partial \mathcal{L}}{\partial \alpha} \right|_{\mathbf{x}=\mathbf{x}^*, \lambda=\lambda^*} = \left. \frac{\partial f}{\partial \alpha} \right|_{\mathbf{x}^*} - \lambda^* \left. \frac{\partial g}{\partial \alpha} \right|_{\mathbf{x}^*}$$

We differentiate the Lagrangian directly with respect to α , holding $\mathbf{x}$ fixed at $\mathbf{x}^*$. The indirect effect through $\mathbf{x}^*(\alpha)$ vanishes by the envelope condition. (*Chiang & Wainwright, p. 394; Simon & Blume, Theorem 19.5*)

Worked Example Envelope Theorem — Consumer Theory

Indirect utility function. For $\max U(x_1, x_2) \text{ s.t. } p_1 x_1 + p_2 x_2 = m$, the value function is the **indirect utility function** $V(p_1, p_2, m)$.

Roy's Identity. By the Envelope Theorem:

$$\frac{\partial V}{\partial m} = \left. \frac{\partial \mathcal{L}}{\partial m} \right|_{\mathbf{x}^*, \lambda^*} = \lambda^*$$

$$\frac{\partial V}{\partial p_i} = \left. \frac{\partial \mathcal{L}}{\partial p_i} \right|_{\mathbf{x}^*, \lambda^*} = -\lambda^* x_i^*$$

Dividing:

$$x_i^*(p, m) = -\frac{\partial V / \partial p_i}{\partial V / \partial m}$$

This is **Roy's Identity** — Marshallian demand is the negative ratio of partial derivatives of the indirect utility function.

Verification with Cobb-Douglas: $V = m / (2\sqrt{p_1 p_2})$. $\partial V / \partial p_1 = -m / (4p_1^{3/2} p_2^{1/2} \cdot \sqrt{p_1 p_2} / p_1) = -m / (4p_1 \sqrt{p_1 p_2}) \dots$

More directly: $\partial V / \partial p_1 = -m \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{p_1 p_2}} \cdot \frac{1}{p_1} = -\frac{m}{4p_1 \sqrt{p_1 p_2} / \sqrt{p_1}}$

Let us compute directly: $V = \frac{m}{2} (p_1 p_2)^{-1/2}$.

$$\frac{\partial V}{\partial p_1} = \frac{m}{2} \cdot \left(-\frac{1}{2}\right) (p_1 p_2)^{-3/2} \cdot p_2 = -\frac{m p_2}{4(p_1 p_2)^{3/2}} = -\frac{m}{4p_1 \sqrt{p_1 p_2}}$$

$$\frac{\partial V}{\partial m} = \frac{1}{2\sqrt{p_1 p_2}} = \lambda^* \quad \checkmark$$

$$\text{Roy's Identity: } x_1^* = -\frac{\partial V / \partial p_1}{\partial V / \partial m} = -\frac{-m / (4p_1 \sqrt{p_1 p_2})}{1 / (2\sqrt{p_1 p_2})} = \frac{m / (4p_1 \sqrt{p_1 p_2})}{1 / (2\sqrt{p_1 p_2})} = \frac{m}{4p_1} \cdot 2 = \frac{m}{2p_1}$$

$\checkmark$

Worked Example Envelope Theorem — Firm's Profit Function

A competitive firm maximises $\pi = Pf(L) - wL - rK$ choosing L (treating P, w, r as parameters).

By the Envelope Theorem (unconstrained version):

$$\frac{d\pi^*}{dP} = \frac{\partial \pi}{\partial P} \Big|_{L=L^*} = f(L^*) = Q^* \geq 0 \quad (\text{Hotelling's Lemma})$$

$$\frac{d\pi^*}{dw} = \frac{\partial \pi}{\partial w} \Big|_{L=L^*} = -L^* \leq 0$$

These results hold *without* solving for L^* explicitly — the envelope theorem gives comparative statics of the value function directly.

8 Quasi-Concavity and Economic Applications

8.1 Quasi-Concavity

Definition Quasi-Concave Function

$f : C \rightarrow \mathbb{R}$ on convex $C \subseteq \mathbb{R}^n$ is **quasi-concave** if for all $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$:

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}$$

Equivalently, all **upper level sets** $\{x \in C : f(x) \geq k\}$ are convex for every k .
 f is **quasi-convex** if $-f$ is quasi-concave.

Quasi-Concavity vs. Concavity

- Concavity $\Rightarrow$ quasi-concavity, but not vice versa. **Example:** $f(x) = x^3$ is quasi-concave (on $\mathbb{R}_+$) but not concave.
- In consumer theory, utility functions need only be **quasi-concave** (not concave) to generate convex indifference curves — which is all that is required for well-behaved demand.
- The FOC characterises the optimum on a quasi-concave objective with a linear constraint under weaker conditions than full concavity.

Theorem Quasi-Concavity and Constrained Optimisation

If f is **quasi-concave** and the constraint set is **convex**, then any critical point of the Lagrangian that satisfies the SOC (bordered Hessian condition) is a **global constrained maximum**. (*Simon & Blume, Theorem 21.10; Hoyer et al., p. 468*)

Worked Example Quasi-Concavity of Cobb-Douglas Utility

$U(x_1, x_2) = x_1^\alpha x_2^\beta$, $\alpha, \beta > 0$, on $\mathbb{R}_{++}^2$.

Upper level sets: $\{(x_1, x_2) : x_1^\alpha x_2^\beta \geq k\}$ for $k > 0$.

Taking logs: $\alpha \ln x_1 + \beta \ln x_2 \geq \ln k$. This is a convex set (intersection of a half-space with $\mathbb{R}_{++}^2$, since $\ln$ is concave). Therefore U is **quasi-concave**.

In fact, U is concave if and only if $\alpha + \beta \leq 1$, but it is quasi-concave for all $\alpha, \beta > 0$. This is why Cobb-Douglas utility generates well-behaved (convex) indifference curves even when $\alpha + \beta > 1$.

9 Comparative Statics in Multi-Variable Models

9.1 The System Approach

In multi-variable models, comparative statics requires differentiating a system of n FOC equations. Let the FOC system be:

$$F_j(x_1^*, \dots, x_n^*; \alpha) = 0, \quad j = 1, \dots, n$$

Totally differentiating and applying the Implicit Function Theorem:

$$\mathbf{J} \begin{pmatrix} dx_1^*/d\alpha \\ \vdots \\ dx_n^*/d\alpha \end{pmatrix} = - \begin{pmatrix} \partial F_1/\partial \alpha \\ \vdots \\ \partial F_n/\partial \alpha \end{pmatrix}$$

where $\mathbf{J}$ is the Jacobian matrix of the FOC system. If $|\mathbf{J}| \neq 0$:

$$\frac{d\mathbf{x}^*}{d\alpha} = -\mathbf{J}^{-1} \frac{\partial \mathbf{F}}{\partial \alpha}$$

Worked Example Comparative Statics — Multiproduct Firm

Return to the monopolist with $\pi = 40Q_1 + 30Q_2 - 3Q_1^2 - Q_1Q_2 - 2Q_2^2$.

FOC system: $\pi_1 = 40 - 6Q_1 - Q_2 = 0$ and $\pi_2 = 30 - Q_1 - 4Q_2 = 0$.

Suppose the demand intercept for good 1 rises to $40 + \alpha$ (a demand shift). The FOC becomes: $40 + \alpha - 6Q_1 - Q_2 = 0$ and $30 - Q_1 - 4Q_2 = 0$.

Differentiating with respect to α :

$$\begin{pmatrix} -6 & -1 \\ -1 & -4 \end{pmatrix} \begin{pmatrix} dQ_1^*/d\alpha \\ dQ_2^*/d\alpha \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

The Jacobian here equals the Hessian $\mathbf{H}$ of π :

$$\begin{pmatrix} dQ_1^*/d\alpha \\ dQ_2^*/d\alpha \end{pmatrix} = -\mathbf{H}^{-1} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \mathbf{H}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathbf{H}^{-1} = \frac{1}{23} \begin{pmatrix} -4 & 1 \\ 1 & -6 \end{pmatrix}$$

$$\Rightarrow dQ_1^*/d\alpha = -4/23 < 0?$$

Wait: $|\mathbf{H}| = (-6)(-4) - (-1)^2 = 24 - 1 = 23$. $\mathbf{H}^{-1} = \frac{1}{23} \begin{pmatrix} -4 & 1 \\ 1 & -6 \end{pmatrix}$

$$\frac{d\mathbf{x}^*}{d\alpha} = \mathbf{H}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{23} \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

So $dQ_1^*/d\alpha = -4/23 < 0$? A demand increase for good 1 reduces optimal output of good 1?

Recheck: $-\mathbf{H}^{-1}(-1, 0)^\top = \mathbf{H}^{-1}(1, 0)^\top$.

The system is $\mathbf{H} \cdot d\mathbf{x} = -\partial\mathbf{F}/\partial\alpha \cdot d\alpha$.

$\partial F_1/\partial\alpha = 1$ (since $F_1 = 40 + \alpha - 6Q_1 - Q_2$), $\partial F_2/\partial\alpha = 0$.

So $\mathbf{H} d\mathbf{x} = -(1, 0)^\top d\alpha$, giving $d\mathbf{x}/d\alpha = -\mathbf{H}^{-1}(1, 0)^\top = -\frac{1}{23}(-4, 1)^\top = (4/23, -1/23)^\top$.

$dQ_1^*/d\alpha = 4/23 > 0$ ✓ (demand rise increases output of good 1).

$dQ_2^*/d\alpha = -1/23 < 0$ (demand rise for good 1 *reduces* output of good 2 — the goods are substitutes in production, since the cross-second derivative $\pi_{12} = -1 < 0$ means producing more Q_1 raises the marginal cost of Q_2).

10 Practice Questions

10.1 Section A: Unconstrained Optimisation

A1. (FOC and Critical Points) Find all critical points of each function.

(a) $f(x, y) = 2x^2 - 4x + y^2 - 4y + 3$

(b) $f(x, y) = x^3 + y^3 - 3xy$

(c) $f(x, y) = (x - 1)^2 + (y - 2)^2 + 3$

(d) $f(x, y, z) = x^2 + y^2 + z^2 - 2x - 4y - 6z$

A2. (Hessian and Classification) For each function below, (i) find all critical points, (ii) compute the Hessian at each critical point, (iii) determine the leading principal minors, and (iv) classify each critical point.

(a) $f(x, y) = 4x^2 + 6xy + 3y^2 - 8x - 6y$

(b) $f(x, y) = x^3 - 3x + y^2 - 2y + 5$

(c) $f(x, y) = e^x(x + y^2)$

A3. (Definiteness and Economic Context) For each Hessian matrix, determine the definiteness and its economic implication for the associated optimisation.

(a) $\mathbf{H} = \begin{pmatrix} -4 & 1 \\ 1 & -3 \end{pmatrix}$

(b) $\mathbf{H} = \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix}$

(c) $\mathbf{H} = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$

(d) $\mathbf{H} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 1 & -2 \end{pmatrix}$

A4. (Multiproduct Firm) A firm produces goods Q_1 and Q_2 with profit:

$$\pi(Q_1, Q_2) = 50Q_1 - 2Q_1^2 + 80Q_2 - 3Q_2^2 - Q_1Q_2 - 100$$

(a) Find the profit-maximising outputs Q_1^* and Q_2^* .

(b) Verify the SOC using the Hessian.

(c) Is the profit function globally concave? Justify.

(d) Find the maximum profit π^* .

- (e) How does Q_1^* change if the marginal cost of good 1 rises by 2 (so the profit becomes $\pi - 2Q_1$)? Use comparative statics.

10.2 Section B: Constrained Optimisation

B1. (Lagrange Method — Basic) Use the Lagrange method to solve each constrained optimisation problem. Verify the bordered Hessian SOC.

- (a) Maximise $f(x, y) = xy$ subject to $x + y = 10$.
- (b) Minimise $f(x, y) = x^2 + y^2$ subject to $x + 2y = 5$.
- (c) Maximise $f(x, y) = x^{1/3}y^{2/3}$ subject to $3x + 2y = 12$.

B2. (Consumer Theory) A consumer has utility $U(x_1, x_2) = \alpha \ln x_1 + (1 - \alpha) \ln x_2$ ($0 < \alpha < 1$) and income $m > 0$ with prices $p_1, p_2 > 0$.

- (a) Set up the Lagrangian and derive the Lagrange conditions.
- (b) Solve for the Marshallian demands $x_1^*(p_1, p_2, m)$ and $x_2^*(p_1, p_2, m)$.
- (c) Find λ^* and interpret it economically.
- (d) Find the indirect utility function $V(p_1, p_2, m)$.
- (e) Use Roy's Identity to recover the Marshallian demands from V .
- (f) Find $\partial x_1^* / \partial p_1$, $\partial x_1^* / \partial p_2$, and $\partial x_1^* / \partial m$. Classify each as own-price, cross-price, and income effects.

B3. (Production and Cost) A firm produces output $Q = K^{1/2}L^{1/2}$ and faces input prices w (wage) and r (rental rate). It wants to minimise cost $C = wL + rK$ subject to producing $\bar{Q}$.

- (a) Derive the conditional factor demands $L^*(w, r, \bar{Q})$ and $K^*(w, r, \bar{Q})$.
- (b) Derive the cost function $C^*(w, r, \bar{Q})$.
- (c) Use Shephard's Lemma ($\partial C^* / \partial w = L^*$ and $\partial C^* / \partial r = K^*$) to verify your factor demands.
- (d) Find the elasticity of substitution between K and L .
- (e) How does the optimal input ratio K^* / L^* change when the relative wage w/r rises?

B4. (Envelope Theorem) A firm maximises $\pi(Q; a) = aQ - Q^2 - c$ where $a > 0$ is a demand parameter and $c > 0$ is a fixed cost (unconstrained, choosing Q).

- (a) Find $Q^*(a)$ and $\pi^*(a)$.
- (b) Use the Envelope Theorem to find $d\pi^* / da$ and $d\pi^* / dc$.
- (c) Verify by direct differentiation of $\pi^*(a) = f(Q^*(a); a)$.
- (d) A government subsidy increases a by 1. What is the gain in maximum

profit? What does this imply for the producer surplus?

B5. (Multiple Constraints — Harder) A consumer maximises $U = x_1^{1/2} + x_2^{1/2} + x_3^{1/2}$ subject to $x_1 + x_2 + x_3 = m$ and $x_1 = x_2$ (a social norm or regulation requiring equal consumption of goods 1 and 2).

- (a) Write the Lagrangian with two multipliers λ and μ .
- (b) Derive all Lagrange conditions.
- (c) Solve for the optimal bundle (x_1^*, x_2^*, x_3^*) .
- (d) Find λ^* and μ^* and interpret each.
- (e) How does the optimal U^* change if m increases by 1?

11 Full Solutions

11.1 Solutions: Section A

Solution

A1. Finding Critical Points

(a) $f(x, y) = 2x^2 - 4x + y^2 - 4y + 3$

$f_x = 4x - 4 = 0 \Rightarrow x^* = 1$; $f_y = 2y - 4 = 0 \Rightarrow y^* = 2$.

One critical point: $(1, 2)$. $f(1, 2) = 2 - 4 + 4 - 8 + 3 = -3$.

(b) $f(x, y) = x^3 + y^3 - 3xy$

$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$; $f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$.

Substituting: $x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$. So $x = 0$ or $x = 1$.

Giving critical points $(0, 0)$ and $(1, 1)$.

$f(0, 0) = 0$; $f(1, 1) = 1 + 1 - 3 = -1$.

(c) $f(x, y) = (x - 1)^2 + (y - 2)^2 + 3$

$f_x = 2(x - 1) = 0 \Rightarrow x^* = 1$; $f_y = 2(y - 2) = 0 \Rightarrow y^* = 2$.

One critical point: $(1, 2)$. $f(1, 2) = 3$ (minimum of a sum of squares, minimum value is 3).

(d) $f(x, y, z) = x^2 + y^2 + z^2 - 2x - 4y - 6z$

$f_x = 2x - 2 = 0$, $f_y = 2y - 4 = 0$, $f_z = 2z - 6 = 0 \Rightarrow (x^*, y^*, z^*) = (1, 2, 3)$.

$f(1, 2, 3) = 1 + 4 + 9 - 2 - 8 - 18 = -14$.

Solution

A2. Hessian and Classification

(a) $f(x, y) = 4x^2 + 6xy + 3y^2 - 8x - 6y$

$f_x = 8x + 6y - 8 = 0$ and $f_y = 6x + 6y - 6 = 0$.

From these: Subtracting: $2x - 2 = 0 \Rightarrow x = 1$; then $6 + 6y - 6 = 0 \Rightarrow y = 0$.

Critical point: $(1, 0)$. $f(1, 0) = 4 - 8 = -4$.

$$\mathbf{H} = \begin{pmatrix} 8 & 6 \\ 6 & 6 \end{pmatrix}. D_1 = 8 > 0, D_2 = 48 - 36 = 12 > 0. \text{ Positive definite: } (1, 0) \text{ is a}$$

local (and global) minimum.

(b) $f(x, y) = x^3 - 3x + y^2 - 2y + 5$

$f_x = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \Rightarrow x = \pm 1$; $f_y = 2y - 2 = 0 \Rightarrow y = 1$.

Two critical points: $(1, 1)$ and $(-1, 1)$.

$$\mathbf{H} = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}.$$

At $(1, 1)$: $D_1 = 6 > 0$, $D_2 = 12 > 0$. **PD**: local minimum. $f(1, 1) = 1 - 3 + 1 - 2 + 5 = 2$.

At $(-1, 1)$: $D_1 = -6 < 0$, $D_2 = -12 < 0$. $D_2 < 0$ means indefinite. **Saddle point.** $f(-1, 1) = -1 + 3 + 1 - 2 + 5 = 6$.

(c) $f(x, y) = e^x(x + y^2)$

$f_x = e^x(x + y^2) + e^x = e^x(x + y^2 + 1) = 0$. Since $e^x > 0$: $x + y^2 + 1 = 0 \Rightarrow x = -1 - y^2$.

$$f_y = e^x \cdot 2y = 0 \Rightarrow y = 0 \text{ (since } e^x > 0 \text{)}.$$

With $y = 0$: $x = -1$. Critical point: $(-1, 0)$. $f(-1, 0) = e^{-1}(-1 + 0) = -e^{-1}$.

$$f_{xx} = e^x(x + y^2 + 2), f_{xy} = 2ye^x, f_{yy} = 2e^x.$$

At $(-1, 0)$: $f_{xx} = e^{-1}(-1 + 0 + 2) = e^{-1}$, $f_{xy} = 0$, $f_{yy} = 2e^{-1}$.

$$\mathbf{H} = \begin{pmatrix} e^{-1} & 0 \\ 0 & 2e^{-1} \end{pmatrix}. D_1 = e^{-1} > 0, D_2 = 2e^{-2} > 0. \text{ PD: local minimum.}$$

Solution

A3. Definiteness and Economic Implication

(a) $D_1 = -4 < 0$, $D_2 = (-4)(-3) - (1)^2 = 12 - 1 = 11 > 0$. Sign pattern $(-, +)$: **Negative definite**. At the corresponding critical point of the objective, this is a **local (and if global, then global) maximum**. In profit maximisation, this confirms the critical output bundle is a profit maximum.

(b) $D_1 = 3 > 0$, $D_2 = 15 - 4 = 11 > 0$. **Positive definite**: critical point is a **local minimum**. In cost minimisation, this confirms the critical input bundle minimises cost.

(c) $D_1 = 2 > 0$, $D_2 = 8 - 9 = -1 < 0$. $D_2 < 0$: **Indefinite**. The critical point is a **saddle point** — neither a maximum nor a minimum.

(d) 3×3 matrix. Check leading principal minors: $D_1 = -2 < 0$, $D_2 =$

$$(-2)(-3) - (0)^2 = 6 > 0, D_3 = \det \begin{pmatrix} -2 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 1 & -2 \end{pmatrix} = -2[(-3)(-2) -$$

$$(1)(1)] = -2[6 - 1] = -10 < 0.$$

Sign pattern $(-, +, -)$: $(-1)^1 D_1 > 0$, $(-1)^2 D_2 > 0$, $(-1)^3 D_3 > 0$. **Negative definite**: critical point is a **local maximum**.

Solution

A4. Multiproduct Firm

$$\pi(Q_1, Q_2) = 50Q_1 - 2Q_1^2 + 80Q_2 - 3Q_2^2 - Q_1Q_2 - 100$$

$$\text{(a) FOC: } \pi_1 = 50 - 4Q_1 - Q_2 = 0 \quad (1) \quad \pi_2 = 80 - Q_1 - 6Q_2 = 0 \quad (2)$$

From (1): $Q_2 = 50 - 4Q_1$. Substitute into (2): $80 - Q_1 - 6(50 - 4Q_1) = 0 \Rightarrow 80 - Q_1 - 300 + 24Q_1 = 0$.

$$\text{Wait: } 80 - Q_1 - 6(50 - 4Q_1) = 80 - Q_1 - 300 + 24Q_1 = 23Q_1 - 220 = 0 \Rightarrow Q_1^* = 220/23.$$

$$Q_2^* = 50 - 4(220/23) = 50 - 880/23 = (1150 - 880)/23 = 270/23.$$

$$\text{(b) Hessian: } \mathbf{H} = \begin{pmatrix} -4 & -1 \\ -1 & -6 \end{pmatrix}. D_1 = -4 < 0, D_2 = 24 - 1 = 23 > 0. \text{ Sign}$$

$(-, +)$: **Negative definite**. Confirmed local (and global) maximum. ✓

(c) Since $\mathbf{H}$ is constant (independent of Q_1, Q_2) and negative definite, π is **strictly concave globally**.

$$\text{(d) } \pi^* = 50(220/23) - 2(220/23)^2 + 80(270/23) - 3(270/23)^2 - (220/23)(270/23) - 100$$

$$\begin{aligned}
&= \frac{1}{23^2} [50(220)(23) - 2(220)^2 + 80(270)(23) - 3(270)^2 - (220)(270)] - 100 \\
&= \frac{1}{529} [253000 - 96800 + 496800 - 218700 - 59400] - 100 \\
&= \frac{374900}{529} - 100 = 708.70\dots - 100 \approx 608.70
\end{aligned}$$

More precisely: $374900/529 = 708.507\dots$, so $\pi^* \approx 608.51$.

(e) Comparative statics. New profit: $\tilde{\pi} = \pi - 2Q_1$, so the FOC for Q_1 becomes: $50 - 2 - 4Q_1 - Q_2 = 0$, i.e. $48 - 4Q_1 - Q_2 = 0$. FOC for Q_2 unchanged.

The Hessian is the same; only the RHS of (1) changes by -2 : $\mathbf{H} \begin{pmatrix} dQ_1 \\ dQ_2 \end{pmatrix} =$

$$\begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} dQ_1^* \\ dQ_2^* \end{pmatrix} = \mathbf{H}^{-1} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \frac{1}{23} \begin{pmatrix} -6 & 1 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \frac{1}{23} \begin{pmatrix} 12 \\ -2 \end{pmatrix}$$

Wait: for $\mathbf{H}$ ND, $|\mathbf{H}| = 23$, and $\mathbf{H}^{-1} = \frac{1}{23} \begin{pmatrix} -6 & 1 \\ 1 & -4 \end{pmatrix}$.

$dQ_1^* = -\mathbf{H}^{-1}(2, 0)^\top$: the MC increase by 2 shifts the FOC by -2 for Q_1 .

Correctly: the perturbation to (1) is -2 (constant shifts the intercept from 50 to 48). The change in FOC: $\Delta\pi_1 = -2$. So $\mathbf{H}(dQ^*) = (2, 0)^\top$ (the negative of the shift in the FOC).

$$dQ^* = \mathbf{H}^{-1}(2, 0)^\top = \frac{1}{23}(-6 \cdot 2 + 1 \cdot 0, 1 \cdot 2 + (-4) \cdot 0)^\top = \frac{1}{23}(-12, 2)^\top$$

$dQ_1^*/d(\text{MC}_1) = -12/23 < 0$: higher MC reduces output of good 1. ✓

$dQ_2^*/d(\text{MC}_1) = 2/23 > 0$: higher MC for good 1 causes the firm to shift output toward good 2.

11.2 Solutions: Section B

Solution

B1. Lagrange Method

(a) $\max xy$ s.t. $x + y = 10$.

$$\mathcal{L} = xy - \lambda(x + y - 10).$$

$$\mathcal{L}_x = y - \lambda = 0, \mathcal{L}_y = x - \lambda = 0, \mathcal{L}_\lambda = x + y - 10 = 0.$$

From the first two: $x = y$. Substituting: $2x = 10 \Rightarrow x^* = y^* = 5$. $\lambda^* = 5$. Max value: $f(5, 5) = 25$.

$$\text{Bordered Hessian: } \bar{\mathbf{H}} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$|\bar{\mathbf{H}}| = 0 - 1(0 - 1) + 1(1 - 0) = 1 + 1 = 2 > 0.$$

For $n = 2, m = 1$ max: need $(-1)^m |\bar{\mathbf{H}}| = (-1)^1 \cdot 2 = -2 < 0$.

Hmm. Actually with this convention: for a max we need $(-1)^m |\bar{\mathbf{H}}| < 0$ where $m = 1$: $-2 < 0$. ✓ Confirmed maximum.

(b) $\min x^2 + y^2$ s.t. $x + 2y = 5$.

$$\mathcal{L} = x^2 + y^2 - \lambda(x + 2y - 5).$$

$$\mathcal{L}_x = 2x - \lambda = 0, \mathcal{L}_y = 2y - 2\lambda = 0, \text{ so } \lambda = 2x \text{ and } \lambda = y.$$

Thus $y = 2x$. Into constraint: $x + 4x = 5 \Rightarrow x^* = 1, y^* = 2, \lambda^* = 2$. Min value: $f(1, 2) = 1 + 4 = 5$.

$$\bar{\mathbf{H}} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

$$|\bar{\mathbf{H}}| = 0 - 1(2 - 0) + 2(0 - 4) = -2 - 8 = -10 < 0.$$

For min with $n = 2, m = 1$: need $(-1)^n |\bar{\mathbf{H}}| = (-1)^2(-10) = -10 < 0$?

Using Chiang's sign condition for min: last $n = 2$ bordered minor has sign $(-1)^{n+m} = (-1)^3 < 0$, so $|\bar{\mathbf{H}}| < 0$ for a min. $|\bar{\mathbf{H}}| = -10 < 0$. ✓ Confirmed minimum.

(c) $\max x^{1/3} y^{2/3}$ s.t. $3x + 2y = 12$.

$$\mathcal{L} = x^{1/3} y^{2/3} - \lambda(3x + 2y - 12).$$

$$\mathcal{L}_x = \frac{1}{3} x^{-2/3} y^{2/3} - 3\lambda = 0 \Rightarrow \frac{y^{2/3}}{3x^{2/3}} = 3\lambda \quad (1)$$

$$\mathcal{L}_y = \frac{2}{3} x^{1/3} y^{-1/3} - 2\lambda = 0 \Rightarrow \frac{x^{1/3}}{3y^{1/3}} = \lambda \quad (2)$$

$$(1)/(2): \frac{y^{2/3}/3x^{2/3}}{x^{1/3}/3y^{1/3}} = 3 \Rightarrow \frac{y}{x} = 3 \Rightarrow y = 3x.$$

Into constraint: $3x + 6x = 12 \Rightarrow x^* = 4/3, y^* = 4$.

$$f^* = (4/3)^{1/3} (4)^{2/3} = 4^{1/3}/3^{1/3} \cdot 4^{2/3} = 4/(3^{1/3}) \approx 4/1.442 \approx 2.77.$$

$$\lambda^* = \frac{x^{*1/3}}{3y^{*1/3}} = \frac{(4/3)^{1/3}}{3 \cdot 4^{1/3}} = \frac{4^{1/3}}{3 \cdot 3^{1/3} \cdot 4^{1/3}} = \frac{1}{3^{4/3}} \approx 0.231.$$

Solution

B2. Consumer Theory with Log Utility

$$U = \alpha \ln x_1 + (1 - \alpha) \ln x_2, \alpha \in (0, 1).$$

$$(a) \text{ Lagrangian: } \mathcal{L} = \alpha \ln x_1 + (1 - \alpha) \ln x_2 - \lambda(p_1 x_1 + p_2 x_2 - m)$$

$$\text{Lagrange conditions: } \mathcal{L}_{x_1} = \alpha/x_1 - \lambda p_1 = 0 \quad (1) \quad \mathcal{L}_{x_2} = (1 - \alpha)/x_2 - \lambda p_2 = 0$$

$$(2) \quad \mathcal{L}_\lambda = p_1 x_1 + p_2 x_2 - m = 0 \quad (3)$$

(b) From (1): $\lambda p_1 x_1 = \alpha \Rightarrow p_1 x_1 = \alpha/\lambda$. From (2): $p_2 x_2 = (1 - \alpha)/\lambda$. Sum: $m = 1/\lambda \Rightarrow \lambda^* = 1/m$.

$$x_1^* = \frac{\alpha}{p_1 \lambda^*} = \frac{\alpha m}{p_1}, \quad x_2^* = \frac{(1 - \alpha)m}{p_2}.$$

(c) $\lambda^* = 1/m$: one extra unit of income raises utility by $1/m$. As income rises, the marginal utility of income falls (consistent with diminishing marginal utility of income).

$$(d) V = \alpha \ln(\alpha m/p_1) + (1 - \alpha) \ln((1 - \alpha)m/p_2)$$

$$= \alpha \ln \alpha + \alpha \ln m - \alpha \ln p_1 + (1 - \alpha) \ln(1 - \alpha) + (1 - \alpha) \ln m - (1 - \alpha) \ln p_2$$

$$= \ln m - \alpha \ln p_1 - (1 - \alpha) \ln p_2 + \alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha)$$

Let $A = \alpha^\alpha (1 - \alpha)^{1-\alpha}$ (a constant). Then: $V(p_1, p_2, m) = \ln m - \alpha \ln p_1 - (1 - \alpha) \ln p_2 + \ln A$.

(e) Roy's Identity:

$$\partial V / \partial p_1 = -\alpha / p_1; \quad \partial V / \partial m = 1/m.$$

$$x_1^* = -\frac{\partial V/\partial p_1}{\partial V/\partial m} = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}. \checkmark$$

(f) $\partial x_1^*/\partial p_1 = -\alpha m/p_1^2 < 0$ (own-price effect: negative, law of demand).

$\partial x_1^*/\partial p_2 = 0$ (cross-price effect: zero, goods are neither substitutes nor complements in Marshallian sense for this separable utility).

$\partial x_1^*/\partial m = \alpha/p_1 > 0$ (income effect: positive, both goods are normal).

Note: the zero cross-price effect is specific to log utility. The Cobb-Douglas form implies zero cross-substitution in this sense.

Solution

B3. Cost Minimisation — Cobb-Douglas

$$\min C = wL + rK \text{ s.t. } K^{1/2}L^{1/2} = \bar{Q}.$$

$$\text{Lagrangian: } \mathcal{L} = wL + rK - \lambda(K^{1/2}L^{1/2} - \bar{Q})$$

$$\text{FOC: } \mathcal{L}_L = w - \frac{\lambda}{2}K^{1/2}L^{-1/2} = 0 \Rightarrow w = \frac{\lambda}{2}\sqrt{K/L} \quad (1)$$

$$\mathcal{L}_K = r - \frac{\lambda}{2}K^{-1/2}L^{1/2} = 0 \Rightarrow r = \frac{\lambda}{2}\sqrt{L/K} \quad (2)$$

(a) Divide (1) by (2): $w/r = K/L \Rightarrow K/L = w/r \Rightarrow K = wL/r$.

Into constraint: $(wL/r)^{1/2}L^{1/2} = \bar{Q} \Rightarrow (w/r)^{1/2}L = \bar{Q} \Rightarrow L^* = \bar{Q}(r/w)^{1/2} = \bar{Q}\sqrt{r/w}$.

$$K^* = \bar{Q}\sqrt{w/r}.$$

(b) $C^* = wL^* + rK^* = w\bar{Q}\sqrt{r/w} + r\bar{Q}\sqrt{w/r} = \bar{Q}(\sqrt{wr} + \sqrt{rw}) = 2\bar{Q}\sqrt{wr}$.

(c) **Shephard's Lemma:** $\partial C^*/\partial w = 2\bar{Q} \cdot \frac{1}{2}\sqrt{r/w} = \bar{Q}\sqrt{r/w} = L^*$. $\checkmark$

$$\partial C^*/\partial r = \bar{Q}\sqrt{w/r} = K^*. \checkmark$$

(d) **Elasticity of substitution:** For $Q = K^{1/2}L^{1/2}$, the elasticity of substitution $\sigma = 1$ (Cobb-Douglas always has unit elasticity).

Formally: $\sigma = \frac{d \ln(K/L)}{d \ln(w/r)} = \frac{d \ln(w/r)}{d \ln(w/r)} = 1$ (since $K/L = w/r$, so $\ln(K/L) = \ln(w/r)$ and the derivative is 1).

(e) $K^*/L^* = w/r$: when the relative wage w/r rises, the firm substitutes capital for labour (uses more K relative to L). This follows directly from cost minimisation — the input ratio equals the relative factor price.

Solution

B4. Envelope Theorem

$$\pi(Q; a, c) = aQ - Q^2 - c.$$

(a) FOC: $a - 2Q = 0 \Rightarrow Q^*(a) = a/2$.

$$\pi^*(a, c) = a(a/2) - (a/2)^2 - c = a^2/2 - a^2/4 - c = a^2/4 - c.$$

(b) **Envelope Theorem:** $d\pi^*/da = \partial\pi/\partial a|_{Q=Q^*} = Q^* = a/2$.

$$d\pi^*/dc = \partial\pi/\partial c|_{Q=Q^*} = -1.$$

(c) **Direct differentiation:** $d\pi^*/da = d(a^2/4 - c)/da = a/2 = Q^*$. $\checkmark$

$$d\pi^*/dc = -1. \checkmark$$

(d) A unit increase in a raises maximum profit by $Q^* = a/2$. This equals the equilibrium output — intuitive since a is the demand intercept and raising it by 1 unit raises price by 1 at every output level, so profit rises by 1 times the quantity sold. This is consistent with Hotelling's Lemma for a competitive firm: $d\pi^*/dP = Q^*$.

Solution**B5. Multiple Constraints**

$\max U = x_1^{1/2} + x_2^{1/2} + x_3^{1/2}$ s.t. $x_1 + x_2 + x_3 = m$ (constraint 1) and $x_1 - x_2 = 0$ (constraint 2).

(a) Lagrangian: $\mathcal{L} = x_1^{1/2} + x_2^{1/2} + x_3^{1/2} - \lambda(x_1 + x_2 + x_3 - m) - \mu(x_1 - x_2)$

(b) Lagrange conditions: $\mathcal{L}_{x_1} = \frac{1}{2}x_1^{-1/2} - \lambda - \mu = 0$ (1)

$$\mathcal{L}_{x_2} = \frac{1}{2}x_2^{-1/2} - \lambda + \mu = 0 \quad (2)$$

$$\mathcal{L}_{x_3} = \frac{1}{2}x_3^{-1/2} - \lambda = 0 \Rightarrow \lambda = \frac{1}{2x_3^{1/2}} \quad (3)$$

$$\mathcal{L}_\lambda = x_1 + x_2 + x_3 - m = 0 \quad (4)$$

$$\mathcal{L}_\mu = x_1 - x_2 = 0 \Rightarrow x_1 = x_2 \quad (5)$$

(c) From (5): $x_1 = x_2$. Let $x_1 = x_2 = a$.

$$\text{Adding (1)+(2): } \frac{1}{2a^{1/2}} + \frac{1}{2a^{1/2}} - 2\lambda = 0 \Rightarrow \frac{1}{a^{1/2}} = 2\lambda.$$

$$\text{From (3): } \lambda = \frac{1}{2x_3^{1/2}}.$$

$$\text{So } \frac{1}{a^{1/2}} = \frac{1}{x_3^{1/2}} \Rightarrow x_3 = a.$$

$$\text{From (4): } a + a + a = m \Rightarrow a = m/3.$$

$$(x_1^*, x_2^*, x_3^*) = (m/3, m/3, m/3).$$

(d) $\lambda^* = \frac{1}{2(m/3)^{1/2}} = \frac{\sqrt{3}}{2\sqrt{m}}$: the marginal utility of income — how much utility rises for a unit increase in m .

$$\text{From (1)-(2): } \mu = \frac{1}{2}x_1^{-1/2} - \lambda - 0 - (\frac{1}{2}x_2^{-1/2} - \lambda) = \frac{1}{2}(x_1^{-1/2} - x_2^{-1/2}) = 0 \text{ (since } x_1 = x_2).$$

$\mu^* = 0$: the social norm constraint $x_1 = x_2$ is *not binding* in the sense that the unconstrained optimum (with just the budget constraint) would also give $x_1 = x_2 = x_3$ by symmetry. The constraint imposes no additional cost on the consumer.

$$\text{(e) } U^* = 3 \cdot (m/3)^{1/2} = 3 \cdot \frac{\sqrt{m}}{\sqrt{3}} = \sqrt{3m}.$$

$$dU^*/dm = \frac{\sqrt{3}}{2\sqrt{m}} = \lambda^*. \quad \checkmark \text{ (Confirms the Envelope Theorem.)}$$

A unit increase in income raises maximum utility by $\sqrt{3}/(2\sqrt{m})$, which decreases as m rises (diminishing marginal utility of income).

Summary: Key Results and Formulae

Result	Statement
FOC (unconstrained)	$\nabla f(\mathbf{x}^*) = \mathbf{0}$; necessary for interior local optimum.
Hessian matrix	$\mathbf{H}_{ij} = \partial^2 f / \partial x_i \partial x_j$; symmetric by Young's Thm.
SOC (unconstrained)	$\mathbf{H}$ ND $\Rightarrow$ local max; $\mathbf{H}$ PD $\Rightarrow$ local min; indefinite $\Rightarrow$ saddle.
Definiteness (ND)	$(-1)^k D_k > 0$ for $k = 1, \dots, n$ (alternating minors).
Definiteness (PD)	$D_k > 0$ for all $k = 1, \dots, n$.
Global optimum	$\mathbf{H}$ ND everywhere $\Rightarrow$ strictly concave $\Rightarrow$ FOC gives unique global max.
Lagrangian	$\mathcal{L} = f - \lambda(g - c)$; FOC gives $n + 1$ equations.
Lagrange multiplier	$\lambda^* = dV/dc$: shadow price of constraint.
Bordered Hessian	$(n + 1) \times (n + 1)$ matrix incorporating gradient of g as border.
SOC (constrained, $n = 2$)	$ \bar{\mathbf{H}} > 0 \Rightarrow$ max; $ \bar{\mathbf{H}} < 0 \Rightarrow$ min (Chiang sign convention).
Envelope Theorem	$dV/d\alpha = \partial \mathcal{L} / \partial \alpha _{\mathbf{x}^*, \lambda^*}$.
Roy's Identity	$x_i^*(p, m) = -(\partial V / \partial p_i) / (\partial V / \partial m)$.
Shephard's Lemma	$\partial C^* / \partial w_i = x_i^*$ (conditional factor demand).
Quasi-concavity	Upper level sets convex; weaker than concavity; sufficient for unique constrained max with convex constraint.
Taylor expansion	$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) \approx \frac{1}{2} \mathbf{d}^\top \mathbf{H}(\mathbf{x}^*) \mathbf{d}$ (when $\nabla f(\mathbf{x}^*) = \mathbf{0}$).

References

- **Primary:** Chiang & Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005. Chapters 11–12.
- **Supplementary:** Hoy et al., *Mathematics for Economics*, 3rd ed., MIT Press, 2011. Chapters 17–19.
- **Supplementary:** Simon & Blume, *Mathematics for Economists*, Norton, 1994. Chapters 16–21.

End of Notes — ECON 320 Optimisation of Multi-Variable Functions — May 2026